

Polarization control by use of the electro-optic effect in periodically poled lithium niobate

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We propose a method to control the polarization of light by the electro-optic effect in periodically poled lithium niobate. A single integrated chip of Z-cut lithium niobate having two sections is used. The first section is not periodically poled, whereas the second section is. With an electric field applied along the Z axis of the first section and another electric field applied along the Y axis of the second section, light with an arbitrary elliptical polarization can be converted into a fixed linearly polarized state. © 2003 Optical Society of America

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1. Introduction

Recently a new artificial nonlinear material, periodically poled LiNbO₃ (PPLN), has received attention owing to its outstanding nonlinear optical properties.^{1–3} Because the nonlinear coefficient changes sign periodically owing to the periodic domain structure of the PPLN, quasi phase matching (QPM) occurs during the frequency conversion process. PPLN is widely used for nonlinear optical interactions, such as second-harmonic generation and optical parametric oscillation.^{4,5} However, besides the nonlinear optical coefficient, the other third-rank tensors, the electro-optic coefficient and the piezoelectric coefficient, are also modulated periodically owing to the periodic domain inversion in PPLN. The electro-optic effect of PPLN has already been studied.⁶ With an electric field applied along the Y axis of Z-cut X-propagation PPLN, coupling between the extraordinary wave and the ordinary wave is established. Under the QPM condition, the polarization of the light propagating in the crystal is modulated by the applied electric field.

This paper is a theoretical study of a device operating as a polarization controller by use of the Jones matrix method and coupled-mode theory. We call

this device the integrated PPLN polarization controller (IPPC). Light with arbitrary elliptical polarization incident on the IPPC can be converted into a fixed linear polarization. The following is the theoretical study of the performance of the IPPC device. We conclude with the theoretical performance and some discussions of such a polarization controller.

2. Principle and Performance Modeling of the Device

The IPPC layout is depicted in Fig. 1. It is a single chip of a LiNbO₃ crystal with two successive sections. The first section has length L_1 with the electric field applied along the Z axis, and the second section is periodically poled with an electric field applied along the Y axis. When an arbitrary polarization state is incident on the crystal, the first section of this device can transform it into linearly polarized light, and the second section of it can transform this linearly polarized light into linearly polarized light of a desired orientation when the two applied electric fields are controlled by a special algorithm.

It is well known that the presence of an external electric field will cause a deformation of the index ellipsoid of an electro-optic birefringent crystal.⁷ In the first section of this device, an electric field is applied along the Z axis. When a light beam is propagating along the x direction, the birefringence induced by the electro-optic effect is

$$\Delta n = n_z - n_y = n_e - n_o - \frac{1}{2}(\gamma_{33}n_e^3 - \gamma_{13}n_o^3)E_z, \quad (1)$$

where E_z is the field intensity, γ_{33} and γ_{13} are the relevant electro-optic coefficients, and n_o and n_e rep-

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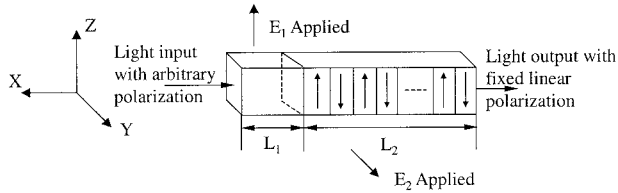


Fig. 1. Schematic diagram of the device. X , Y , and Z represent the principal axes of the original index ellipsoid. The arrows inside the PPLN indicate the spontaneous polarization directions.

represent the refractive indices of the ordinary and extraordinary waves, respectively.

We express the input light in terms of the normalized Jones vector

$$A_{\text{in}} = \begin{pmatrix} \cos \phi \\ \sin \phi \exp(i\delta) \end{pmatrix}. \quad (2)$$

When an electric field is applied along the z direction, the Jones matrix of the first part with length L_1 is described as follows:

$$J = \begin{bmatrix} \exp\{-i\pi[n_e - n_o - \frac{1}{2}(r_{33}n_e^3 - r_{13}n_o^3)E_z]L_1/\lambda\} & 0 \\ 0 & \exp\{i\pi[n_e - n_o - \frac{1}{2}(r_{33}n_e^3 - r_{13}n_o^3)E_z]L_1/\lambda\} \end{bmatrix}. \quad (3)$$

Thus the light emitted from the first section is described as

$$\begin{aligned} A_{\text{out1}} &= JA_{\text{in}} = \begin{bmatrix} \exp\{-i\pi[n_e - n_o - \frac{1}{2}(r_{33}n_e^3 - r_{13}n_o^3)E_z]L_1/\lambda\} & 0 \\ 0 & \exp\{i\pi[n_e - n_o - \frac{1}{2}(r_{33}n_e^3 - r_{13}n_o^3)E_z]L_1/\lambda\} \end{bmatrix} \\ &\times \begin{pmatrix} \cos \phi \\ \sin \phi \exp(i\delta) \end{pmatrix} \\ &= \begin{pmatrix} \cos \phi \exp\{-i\pi[n_e - n_o - \frac{1}{2}(r_{33}n_e^3 - r_{13}n_o^3)E_z]L_1/\lambda\} \\ \sin \phi \exp[i\delta + i\pi[n_e - n_o - \frac{1}{2}(r_{33}n_e^3 - r_{13}n_o^3)E_z]L_1/\lambda] \end{pmatrix}. \end{aligned} \quad (4)$$

When $2\pi[n_e - n_o - \frac{1}{2}(r_{33}n_e^3 - r_{13}n_o^3)E_z]L_1/\lambda + \delta = m\pi$ ($m = \pm 1, \pm 2, \dots$), the input light with arbitrary elliptical polarization will be transformed into linearly polarized light. Figure 2 shows the relationship between the applied electric field and the relative phase delay between the extraordinary and the ordinary waves of input light with the crystal lengths of 1.031 and 0.5155 cm, respectively. From Fig. 2, when the phase delay between the extraordinary and the ordinary waves of the input light is from $-\pi$ to 0, a 0–0.7 kV/mm electric field is needed to transform an elliptical polarization into a linear polarization with a crystal length equal to 1.031 cm.

The second section of the crystal is periodically poled. With an external electric field applied along

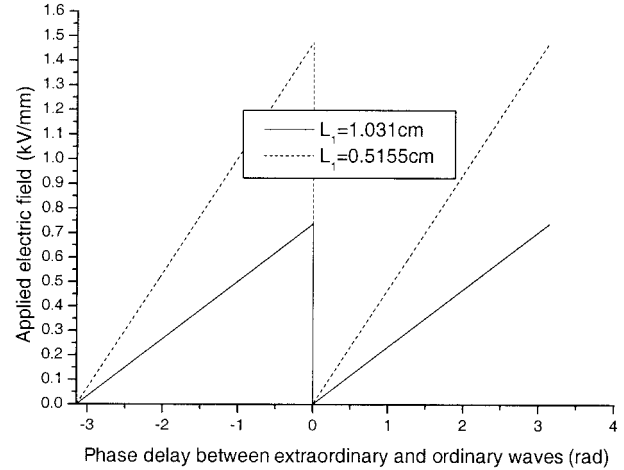


Fig. 2. Relationship between the applied electric field and the relative phase delay between the extraordinary and the ordinary waves for the first section of the device.

the Y axis, the index ellipsoid deforms to make the Y and Z axes rotate a small angle about the X axis^{6,7}:

$$\theta \approx \frac{\gamma_{51}E_y}{(1/n_e^2) - (1/n_o^2)}, \quad (5)$$

where E_y is the field intensity, γ_{51} is the electro-optic coefficient, and n_o and n_e represent the refractive indices of the ordinary and extraordinary waves, respectively.

In PPLN all elements of the electro-optic tensors have different signs in different domains.⁶ After a field is applied along the Y axis, the azimuth angle of the new Z axis and Y axis rotates right and left from $+\theta$ to $-\theta$ successively owing to the periodic electro-

optic coefficient change in PPLN. When the domain thickness satisfies the phase-matching condition, each domain acts as a half-wave plate. Thus we can take this structure as a folded Solc-type filter. As a result, polarization of light propagating in PPLN can be rotated in the presence of the external applied electric field.⁶

The coupled-mode theory can be used to study such a structure. In PPLN the coupled-wave equations of the ordinary and extraordinary waves are

$$\begin{aligned} dA_1/dx &= -i\kappa A_2 \exp(i\Delta\beta x), \\ dA_2/dx &= -i\kappa^* A_1 \exp(-i\Delta\beta x), \end{aligned} \quad (6)$$

with $\Delta\beta = (\beta_2 - \beta_1) - G_m$, $G_m = 2\pi m/\Lambda$, and

$$\begin{aligned} \kappa &= -\frac{\omega}{2c} \frac{n_o^2 n_e^2 \gamma_{51} E_y}{\sqrt{n_o n_e}} \frac{i(1 - \cos m\pi)}{m\pi} \\ (m &= 1, 3, 5, \dots), \end{aligned}$$

where A_1 and A_2 are the normalized amplitudes of the ordinary wave and the extraordinary wave, respectively, β_1 and β_2 are the corresponding wave vectors, G_m is the m th reciprocal vector corresponding to the periodicity of poling, and Λ is the period of the PPLN that is equal to twice the domain thickness if the duty cycle is 50%. The boundary condition of the equation is given by

$$A_{\text{in}2} = A_{\text{out}1} = \begin{pmatrix} \cos \phi \\ \exp(-im\pi) \sin \phi \end{pmatrix} = \begin{pmatrix} \cos \phi_1 \\ \sin \phi_1 \end{pmatrix}, \quad (7)$$

where

$$\phi_1 = \begin{cases} \phi & m = 0, \pm 2, \dots \\ -\phi & m = \pm 1, \pm 3, \dots \end{cases} \quad (-\pi < \phi_1 < \pi).$$

So the solution of coupled-mode equation (6) is

$$\begin{aligned} A_1(x) &= \exp[i(\Delta\beta/2)x] \left[\left(\cos sx - i \frac{\Delta\beta}{2s} \sin sx \right) \cos \phi_1 \right. \\ &\quad \left. - i \frac{\kappa}{s} \sin sx \sin \phi_1 \right], \end{aligned} \quad (8)$$

$$\begin{aligned} A_2(x) &= \exp[-i(\Delta\beta/2)x] \left[-i \frac{\kappa^*}{s} \sin sx \cos \phi_1 \right. \\ &\quad \left. + \left(\cos sx + i \frac{\Delta\beta}{2s} \sin sx \right) \sin \phi_1 \right], \end{aligned} \quad (9)$$

where s is given by $s^2 = \kappa^* \kappa + (\Delta\beta/2)^2$.

When $\Delta\beta = (\beta_2 - \beta_1) - G_m = 0$, the reciprocal vector compensates for the wave-vector mismatch. As in QPM frequency conversion, we call this condition a QPM polarization conversion condition. If the coherence length is defined as $L_c = \lambda/2(n_o - n_e)$, the QPM condition is satisfied for light with wavelength λ if each domain thickness is L_c or its multiple. Under such a QPM condition, the normalized amplitudes of the ordinary wave and the extraordinary

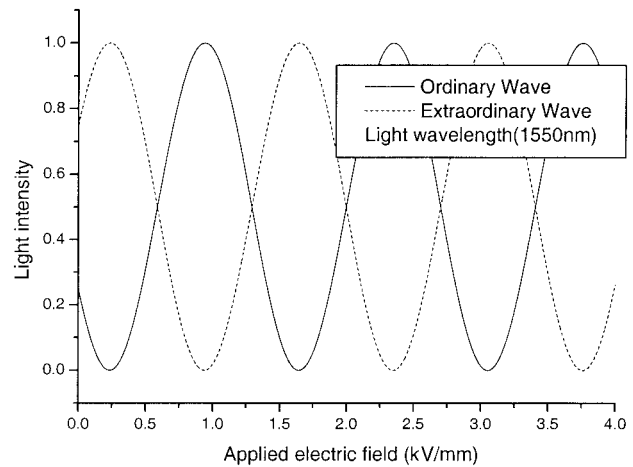


Fig. 3. Power exchange relations between the extraordinary and the ordinary waves when the QPM condition is satisfied ($\Delta\beta = 0$).

wave A_1 and A_2 at the output of the PPLN section are given by

$$\begin{aligned} A_1(L_2) &= \cos sL_2 \cos \phi_1 - \sin sL_2 \sin \phi_1 \\ &= \cos(sL_2 + \phi_1), \end{aligned} \quad (10)$$

$$\begin{aligned} A_2(L_2) &= \sin sL_2 \cos \phi_1 + \cos sL_2 \sin \phi_1 \\ &= \sin(sL_2 + \phi_1). \end{aligned} \quad (11)$$

When the applied electric field is carefully controlled so that the equation $sL_2 + \phi_1 = (2k + 1)\pi/2$, i.e., $|\kappa|L_2 + \phi_1 = (2k + 1)\pi/2$, is satisfied, then

$$\begin{aligned} A_1(L_2) &= 0, \\ A_2(L_2) &= 1. \end{aligned}$$

Thus the output light is linearly polarized along the Y axis. Figure 3 shows the power exchange relations between the extraordinary and the ordinary waves when the QPM condition is satisfied. For a sample with a domain thickness of $10.31 \mu\text{m}$, the first-order QPM condition corresponds to a wavelength of 1550 nm. The conversion of the arbitrary linear polarization light into the desired linear polarization requires that a certain electrical field be applied along the Y axis of the PPLN section. Figure 4 shows the relationship between the electric field applied to the PPLN section with a length of 1.031 cm and the polarization angle of the linearly polarized light emitted from the first section of the device.

When the polarization state of the input light to the IPPC is known, we can determine the applied electric fields for the first and the second sections from Figs. 2 and 4. The maximum applied electric field is approximately 0.7 kV/mm when both of the two sections are 1.031 cm in length. As a result, input light with arbitrary elliptical polarization can be converted into a fixed linear polarization when the two applied electrical voltages in the IPPC are controlled.

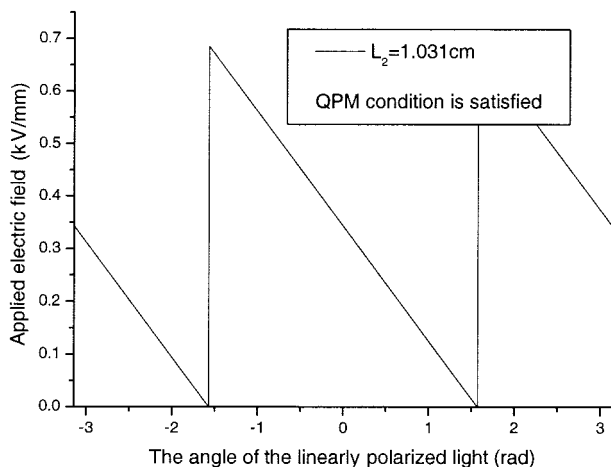


Fig. 4. Relationship between the electric field applied to the PPLN section and the polarization angle of the linearly polarized light emitted from the first section of the device.

3. Conclusion

We propose a integrated PPLN polarization controller that can transform the polarization of light by the application of external fields on a two-section, single chip of a Z-cut LiNbO_3 crystal. When the QPM condition for light with wavelength λ is satisfied, the incident light with arbitrary elliptical polarization can be transformed into light with a fixed linear polarization by the electro-optic effects of the LiNbO_3 crystal and the PPLN crystals. This device can be used as a electrical polarization controller in optical

communication systems. Furthermore, the IPPC will be attractive and practical when such a device is made of waveguide PPLN because only a low operating voltage is required.

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